

Legal Ease

Application Architecture Overview

Legal Ease turns dense legal documents (contracts, leases, terms of service) into plain-language explanations, with every explanation tied back to the exact clause it came from.

Layer	Purpose	Suggested Tech
1. Client	Upload/scan documents, view highlights, chat Q&A	React Native or Flutter; Next.js web
2. API	Auth, routing, rate limiting, core services	API gateway, JWT/OAuth, FastAPI or Node/TypeScript
3. Document pipeline	Ingest, OCR, extract, chunk, embed	Object storage, OCR engine, async workers
4. AI layer	Summaries, clause explanations, risk flags	LLM + retrieval (pgvector), guardrails
5. Data	Users, documents, analyses, embeddings, cache	PostgreSQL, vector store, Redis
6. Security	Protect sensitive documents; compliance	Encryption, isolation, audit logs
7. Infrastructure	Deploy, scale, observe	Containers, CI/CD, monitoring

1. Client Layer

- Mobile app (React Native or Flutter) and web app (Next.js).
- Features: document upload and scan, clause highlighting, Q&A chat, saved history.

2. API Layer

- API gateway with authentication (OAuth/JWT), rate limiting, and logging.
- Backend services: users, documents, analysis, and billing.

3. Document Pipeline

- Upload to encrypted object storage.
- OCR and text extraction for scans and PDFs.
- Chunk by clause/section, then generate embeddings.

4. AI Layer

- LLM for summarizing, explaining clauses, and flagging risky terms.
- Retrieval so every answer cites the actual document text.
- Guardrails: 'not legal advice' disclaimers, confidence flags, no invented citations.

5. Data Layer

- PostgreSQL for users, documents, and analyses.
- Vector store for embeddings (e.g., pgvector).
- Redis for caching and job queues.

6. Security and Compliance

- Encryption at rest and in transit; per-user data isolation.
- Configurable retention and delete-on-request.
- Audit logs; GDPR/CCPA alignment.

7. Infrastructure

- Containerized services on a cloud provider.
- Async workers for heavy processing.
- CI/CD pipeline and monitoring/alerting.

Key Design Principle

Because users rely on the output, every explanation must point back to the exact clause it came from. Legal Ease provides information, not legal advice; high-stakes matters should be reviewed by a licensed attorney.